

- 15** A point source of sound emits energy equally in all directions at a constant rate and a detector placed 2.0 m from the source measures an intensity of 3.0 W m^{-2} . The amplitude at the source is then doubled.

What intensity would the detector measure if it is now placed at a distance 4.0 m from the source?

A 3.0 W m^{-2}

B 1.5 W m^{-2}

C 0.75 W m^{-2}

D 0.38 W m^{-2}

Ans: A

$$I \propto 1/r^2$$

$$I_1 \propto A^2$$

$$3.0 = k/(2.0)^2 \rightarrow k = 12$$

$$I_2 \propto (2A)^2 = 4I_1$$

$$I_2 = 4I_1 = 4[12/(4.0)^2] = 3.0 \text{ W m}^{-2}$$

Option B: used $I \propto 1/r$

Option C: used $I \propto A$ and $I \propto 1/r$